Contents

1 Appendix B: Environmental Channel Modeling

1.1 Objective

Comprehensive atmospheric channel modeling validated against peer-reviewed atmospheric physics: molecular absorption (HO, CO, CH, O), Rayleigh scattering, aerosol effects, channel capacity mapping with 8-14 μm window emphasis, wavelength optimization for 10-100 m ranges, and environmental sensitivity analysis for IR communication in insect olfactory systems.

1.2 Methods (src)

- `src/case_studies/environmental_channel.py`
  - `molecular_absorption_cross_section(wavelengths, molecule_type)` — H2O, CO2, CH4 absorption
  - `rayleigh_scattering_coefficient(wavelengths, air_density)` — molecular scattering
  - `atmospheric_transmission_comprehensive(wavelengths, conditions)` — multi-component transmission
  - `channel_capacity_analysis(wavelengths, environmental_conditions)` — Shannon capacity mapping
  - `optimize_wavelength_for_range(target_range, capacity_requirements)` — wavelength selection
  - `environmental_sensitivity_analysis(parameter_variations)` — parameter sensitivity
  - `atmospheric_transmission_detailed(wavelengths, humidity, temperature, path)` — basic transmission utility
  - `channel_capacity_vs_env(material_props, env_grid)` — grid mapping of capacity vs environment

1.3 Script and outputs

- Script: `scripts/generate_environmental_channel_analysis.py`
- Data: `output/data/environmental_channel_comprehensive.npz`
- Figure: `figures/environmental_channel_comprehensive_analysis.png`

[width=0.9]figures/environmental$_{c}hannel_{c}omprehensive_{a}nalysis.png$

Figure 1: *Comprehensive environmental channel outputs produced deterministically by* ‘$scripts/generate_{e}nvironmental_{c}hannel_{a}nalysis.py$‘$using$‘$src/case_{s}tudies/environmental_{c}hannel.py$‘$.Panelsshowmolec$

[width=0.9]figures/environmental$_{c}hannel_{c}omprehensive_{a}nalysis.png$

Figure 2: *Environmental channel analysis generated by* ‘$scripts/generate_{e}nvironmental_{c}hannel_{a}nalysis.py$‘$.Panelsshowatmospherictransmission, capacitymapping, SNR, and$

1.4 Figure

1.5 Equation references

- Atmospheric transmission: see (**??**)
- Channel capacity: see (**??**)

1.6 Reproducibility

- Run: `python3 scripts/generate_environmental_channel_analysis.py`
- Artifacts saved to `output/data/` and `figures/`.
- Deterministic grids via `src/config.set_random_seed(42)`.

1.7 Context Note on Biological Ranges

Some insects exhibit sensitivity to thermal IR in natural behaviors (e.g., Aedes aegypti up to ~0.7 m; Chandel et al. 2024). These thermal constraints complement our electromagnetic window analysis and motivate species- and wavelength-specific range predictions.

1.8 Cross-references

- Methods: sec:methodology
- Symbols: sec:symbols$_{g}lossaryMathappendix : sec : mathematical_{a}ppendix$

[width=0.9]figures/integrated$_a nalysis_i nformation_a nalysis.png$

Figure 3: *Integrated information analysis (from*
‘$scripts/generate_i ntegrated_a nalysis.py$‘$)showingmolecular, receptor, neuralandenvironmentalinformationdecompositio$